

ALK fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes ALK gene fusions from the msk_impact_50k_2026 study, covering 272 fusion events. 226/272 fusions (83.1%) were in-frame. 261/272 fusions (96.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test ($p=0.00191661$, statistically significant at $\alpha=0.05$). No literature benchmark is configured for ALK.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 272 analyzed fusion events, identifying 32 distinct fusion partner genes. The most frequent partner was EML4, observed in 221 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 222/226 (98.2%) of in-frame fusions retained the domain, $p=0.00191661$ (statistically significant at $\alpha=0.05$). The Protein kinase domain appears to be required for retention. A breakpoint-position permutation test produced a corroborating empirical p-value of 0.000999001.

Domain disruption

Domain-disruption analysis was skipped: ALK has no disruption_required_domains configured; domain-disruption analysis was skipped.

Cutpoint detection

Cutpoint detection scanned 271 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 972 aa (permutation-corrected $p=0.000999001$, statistically significant at $\alpha=0.05$). This is 12 aa from the nearest configured domain boundary at 960 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=261$) and "not_retained" ($n=10$). For Frame_status, retained: 222/227 (97.8%, 95% CI 94.9%-99.1%); not_retained: 4/5 (80.0%, 95% CI 37.6%-96.4%). Welch's t-test comparing Tumor_variant_count between groups gave means of 65.6139 vs 31.8, $p=0.00432792$ (statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: ALK has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	221	0.8125	0.30004340774793187	None	0.8865878486549185	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.6503541781991878	1
LRP1B	1	0.003676470588235294	0.30004340774793187	None	1.0	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3683101330469789	2
EMILIN1	1	0.003676470588235294	0.30004340774793187	None	0.9559412550066756	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.36004911836073056	3
CLTC	2	0.007352941176470588	0.30004340774793187	None	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3569217868206379	4
FN1	1	0.003676470588235294	0.30004340774793187	None	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35554311035004965	5
NOTCH1	1	0.003676470588235294	0.30004340774793187	None	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35554311035004965	6
NRP2	1	0.003676470588235294	0.30004340774793187	None	0.931909212283044	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35554311035004965	7
STRN	7	0.025735294117647058	0.30004340774793187	None	0.8865153538050734	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35530382070895955	8
KIF5B	5	0.01838235294117647	0.30004340774793187	None	0.8945260347129506	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3540484704380101	9
PLEKHA7	1	0.003676470588235294	0.30004340774793187	None	0.9158878504672897	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3525391050095957	10
SPTBN1	4	0.014705882352941176	0.30004340774793187	None	0.8895193591455274	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.35173104229853	11

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
RANBP2	3	0.011029411764705883	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34953878104823544	12
PICALM	3	0.011029411764705883	0.30004340774793187	None	0.8811748998664887	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.348787779713122	13
SQSTM1	2	0.007352941176470588	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34816010457764723	14
TPM1	2	0.007352941176470588	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.34816010457764723	15
CTNNA1	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	16
DCTN1	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	17
DIAPH2	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	18
DYSF	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	19
GCC2	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	20
HMBOX1	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	21
PLEKHH2	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	22
PPP1CB	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	23
TFG	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	24

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
TRAF3	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	25
TTC27	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	26
VCL	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	27
ZFPM2	1	0.003676470588235294	0.30004340774793187	None	0.8851802403204272	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.346781428107059	28
H6PD	1	0.003676470588235294	0.30004340774793187	None	0.7503337783711616	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.3214977164915717	29
CTSE	1	0.003676470588235294	0.30004340774793187	None	0.39118825100133514	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.2541579301097292	30
SOS1	1	0.003676470588235294	0.30004340774793187	None	None	0.6853431332412958	['confidence_certainty', 'domain_retention', 'recurrence']	0.2225355483655125	31
ASAP2	1	0.003676470588235294	0.30004340774793187	None	0.0	0.6853431332412958	['confidence_certainty', 'cutpoint_proximity', 'domain_retention', 'recurrence']	0.1808101330469789	32

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
223	0	261	1	9	0.0	0.03690036900369004	1.4329692908744058
516	0	261	2	8	0.0	0.001230012300123001	2.9100905455940684
785	0	261	3	7	0.0	3.658029145347215e-05	4.436752838604533
939	2	259	3	7	0.018018018018018018	0.0003515999093716071	3.4539512455419974
972	2	259	4	6	0.011583011583011582	1.3820564716971664e-05	4.8594742110534295
1023	10	251	4	6	0.05976095617529881	0.0007961806348620647	3.098988389769786
1028	11	250	4	6	0.066	0.0010657190672512412	2.9723572639982643
1032	12	249	4	6	0.07228915662650602	0.0013947454739763106	2.8555050392241372

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
1035	13	248	4	6	0.07862903225806452	0.0017901726849544938	2.747105073765603
1040	14	247	4	6	0.08502024291497975	0.002258991897970247	2.646085326703862
1042	16	245	4	6	0.09795918367346938	0.003444927806265043	2.4628198749665864
1045	18	243	4	6	0.11111111111111111	0.005008607520231548	2.300282998750876
1047	19	242	4	6	0.11776859504132231	0.005949364209574388	2.225529443519691
1051	20	241	4	6	0.12448132780082988	0.007005060123461933	2.154588132872754
1052	22	239	4	6	0.13807531380753138	0.00948736346530418	2.0228544610771384
1058	254	7	10	0	0.0	1.0	-0.0
1059	255	6	10	0	0.0	1.0	-0.0
1061	258	3	10	0	0.0	1.0	-0.0
1064	259	2	10	0	0.0	1.0	-0.0
1070	260	1	10	0	0.0	1.0	-0.0

domain_retention tables

frame_domain_contingency_table

222	39
4	6

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	271	261	0	10	0.0

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
20	254	272	0.9338235294117647

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ASAP2	1

Partner_gene	Event_count
CLTC	2
CTNNA1	1
CTSE	1
DCTN1	1
DIAPH2	1
DYSF	1
EMILIN1	1
EML4	221
FN1	1
GCC2	1
H6PD	1
HMBOX1	1
KIF5B	5
LRP1B	1
NOTCH1	1
NRP2	1
PICALM	3
PLEKHA7	1
PLEKHH2	1
PPP1CB	1
RANBP2	3
SOS1	1
SPTBN1	4
SQSTM1	2
STRN	7
TFG	1
TPM1	2
TRAF3	1
TTC27	1
VCL	1
ZFPM2	1

Per-event results (results.tsv)

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0024 679-T01-IM 6-39	P-0024 679-T0 1-IM6		ALK- CTS E	CTS E	unk now n	five _pr ime	295435 94	7	516	True	lost	0.0	False		Prot ein k inas e do mai n		ALK (NM_004304) - CTSE(NM_001910) Rearrangement: t(1;2)(q32. 1;p23.3)(chr1:g.206318648 ; chr2:g.29543594) Antisense Fusion	NA
EVT-P-0020 387-T01-IM 6-40	P-0020 387-T0 1-IM6		ALK- DCT N1	DCT N1	in-fr ame	thre e _pr ime	294468 96	20	1058	True	retai ned	1.0	False	Protein kinase domai n			DCTN1 (NM_004082) - ALK (NM_004304) rearrangement: c.3196+23 8:DCTN1_c.3173-502:ALK del Protein Fusion: in frame {DCTN1:ALK}	NA
EVT-P-0006 776-T01-IM 5-44	P-0006 776-T0 1-IM5		ALK- DYS F	DYS F	unk now n	five _pr ime	294474 18	19	1058	True	lost	0.0	False		Prot ein k inas e do mai n		ALK (NM_004304) - DYSF (NM_001130987) rearrangement: c.3172+90 9:ALK_c.3402+7348:DYSF dup Antisense fusion	NA
EVT-P-0031 827-T02-IM 6-45	P-0031 827-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thre e _pr ime	294466 46	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490- 639:EML4_c.3173-252:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028 287-T01-IM 6-46	P-0028 287-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e _pr ime	294466 61	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.4 94-2623:EML4_c.3173-267 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009 301-T01-IM 5-47	P-0009 301-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e _pr ime	294466 51	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.668-6693: EML4_c.3173-257:ALKinv Protein fusion: in frame {EML4-ALK}	NA
EVT-P-0009 301-T02-IM 6-48	P-0009 301-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thre e _pr ime	294466 51	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.668-6 693:EML4_c.3173-257:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009 339-T01-IM 5-49	P-0009 339-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e _pr ime	294478 89	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.6 68-6870:EML4_c.3172+43 8:ALKinv Protein fusion: in frame {EML4-ALK}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ ro le	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_tru ncated	retain ed_do mains	lost _do mai ns	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0009 339-T02-IM 5-50	P-0009 339-T0 2-IM5		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294478 89	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.6 68-6870:EML4_c.3172+43 8:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009 343-T01-IM 5-51	P-0009 343-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294465 21	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.6 68-1093:EML4_c.3173-127 :ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0009 353-T01-IM 5-52	P-0009 353-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294464 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) : c.6 67+2407:EML4_c.3173-71: ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0031 406-T01-IM 6-53	P-0031 406-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294464 25	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 with ALK exons 20-29): c.668-2889: EML4_c.3173-31:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 170-T01-IM 6-54	P-0032 170-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294476 95	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1822:EML4_c.3172+632 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0010 041-T01-IM 5-55	P-0010 041-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294479 42	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fuses with ALK exons 20-29) : c.7 01-6977:EML4_c.3172+38 5:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0008 332-T01-IM 5-56	P-0008 332-T0 1-IM5		ALK- EML 4	EML 4	unk now n	five _pr ime	294469 12	20	1058	True	lost	0.0	False		Prot ein k inas e do mai n		EML4 (NM_019063) - ALK (NM_004304) rearrangement : c.1489+82 8:EML4_c.3173-518:ALKd up Antisense fusion	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0010 177-T01-IM 5-57	P-0010 177-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294473 58	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29): c.1490-2649:EML4_c.3173-964:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0007 705-T01-IM 5-58	P-0007 705-T0 1-IM5		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294474 17	19	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29): c.1594:EML4_c.3172+910:ALKinv Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0010 978-T01-IM 5-59	P-0010 978-T0 1-IM5		ALK- EML 4	EML 4	out- of-fr ame	thr_ee _pr ime	294464 14	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0010 978-T02-IM 5-60	P-0010 978-T0 2-IM5		ALK- EML 4	EML 4	out- of-fr ame	thr_ee _pr ime	294464 14	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.2154+2237:EML4_c.3173-20:ALKinv Protein fusion: out of frame (EML4-ALK)	NA
EVT-P-0011 018-T01-IM 5-61	P-0011 018-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294483 14	19	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011 203-T01-IM 5-62	P-0011 203-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 05	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29) Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011 203-T02-IM 5-63	P-0011 203-T0 2-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 96	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-6805:EML4_c.3173-702:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0011 203-T03-IM 6-64	P-0011 203-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 55	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 5 fused to ALK exon 20): c.494-6851:EML4_c.3173-661:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0012 679-T01-IM 5-65	P-0012 679-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 66	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1489+2812:EML4_c.3172+861:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012 778-T01-IM 5-66	P-0012 778-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 42	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667+1320:EML4_c.3172+885:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0012 778-T02-IM 6-67	P-0012 778-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 42	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+1320:EML4_c.3172+885:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0007 211-T01-IM 5-68	P-0007 211-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 89	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29) : c.667+165: EML4_c.3172+838:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0007 125-T01-IM 5-69	P-0007 125-T0 1-IM5		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294465 97	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-14 with ALK exons 20-29) : c.1328: EML4_c.3173-203:ALKinv Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0006 950-T02-IM 6-70	P-0006 950-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294473 84	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons20-29): c.1489+2302:EML4_c.3172+943:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0013 202-T01-IM 5-71	P-0013 202-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 78	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+2565: EML4_c.3173-784:ALKinv Protein fusion: in frame (EML4-ALK)	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ role	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0006 950-T01-IM 5-72	P-0006 950-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294473 84	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1489+2302:EML4_c.3172+943: ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013 453-T01-IM 5-73	P-0013 453-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294465 97	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+551:EML4_c.3173-203:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013 864-T01-IM 5-74	P-0013 864-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294482 09	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1489+658:EML4_c.3172+118 :ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0013 869-T01-IM 5-75	P-0013 869-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294467 12	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1 to 13 with exons 20-29 of ALK) : c.1490-187:EML4_c.3173-318: ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014 270-T01-IM 6-76	P-0014 270-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294466 79	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused in-frame to ALK exons 20-29) : c.667+2294:EML4_c.3173-285:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0014 310-T01-IM 6-77	P-0014 310-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294475 44	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20- 29): c.667+6128:EML4_c.3172+783:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006 370-T01-IM 5-78	P-0006 370-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thre e_ _pr ime	294481 24	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 with ALK exons 20-29) : c.1490-2147:EML4_c.3172+203:ALKinv Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0014 385-T01-IM 6-79	P-0014 385-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294476 76	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1890:EML4_c.3172+651:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0006 313-T01-IM 5-80	P-0006 313-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294479 66	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) -ALK (NM_004304) Fusion (EML4 exons 1 to 18 fused with exons 20 to 29 of ALK) Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0030 590-T01-IM 6-81	P-0030 590-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294477 04	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.494-5628:EML4_c.3173-350:ALKinv. Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0000 474-T01-IM 3-82	P-0000 474-T0 1-IM3		ALK- EML 4	EML 4	in-frame	three_prime	294470 02	20	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0000 557-T01-IM 3-83	P-0000 557-T0 1-IM3		ALK- EML 4	EML 4	in-frame	three_prime	294465 43	20	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0000 732-T01-IM 3-84	P-0000 732-T0 1-IM3		ALK- EML 4	EML 4	in-frame	three_prime	294475 37	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015 197-T01-IM 6-85	P-0015 197-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294480 09	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6 fused to ALK exon 20) : c.668-2108:EML4_c.3172+318:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0015 434-T02-IM 6-86	P-0015 434-T0 2-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294481 97	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 13 fused to ALK exon20): c.1489+63:EML4_c.3172+130:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0005 321-T01-IM 5-87	P-0005 321-T0 1-IM5		ALK- EML 4	EML 4	in-frame	three_prime	294482 50	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 13 fused with ALK exon 20): c.2068+156 :EML4_c.3173-693:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0029 971-T01-IM 6-88	P-0029 971-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294478 07	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -119:EML4_c.3172+520:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 921-T03-IM 6-89	P-0015 921-T0 3-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294468 09	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused with ALK exons 20-29): c.6 67+516:EML4_c.3173-415: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T01-IM 6-90	P-0015 961-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294465 26	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon 6fused to ALK exon 20): c.668-6666:EML 4_c.3173-132:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T02-IM 6-91	P-0015 961-T0 2-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294465 26	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6 666:EML4_c.3173-132:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T03-IM 6-92	P-0015 961-T0 3-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294465 26	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.668-6 666:EML4_c.3173-132:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 827-T01-IM 6-93	P-0031 827-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294467 00	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-673:EML4_c.3173-306:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027 119-T01-IM 6-94	P-0027 119-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294467 11	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused to ALK exons 20-29): c.1490- 837:EML4_c.3173-317:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_n_t_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0029 750-T01-IM 6-95	P-0029 750-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr ee _pr ime	294466 60	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490-2574:EML4_c.3173-266:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0029 550-T01-IM 6-96	P-0029 550-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr ee _pr ime	294478 63	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.494-721:EML4_c.3172+464:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016 280-T01-IM 6-97	P-0016 280-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr ee _pr ime	294479 05	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) rearrangement: c.1316-2732:EML4_c.3172+422:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0016 580-T01-IM 6-98	P-0016 580-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr ee _pr ime	294475 79	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+2257:EML4_c.3172+748:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0005 305-T01-IM 5-99	P-0005 305-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr ee _pr ime	294477 64	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.668-5757:EML4_c.3172+563:ALKinv Protein fusion: in frame {EML4:ALK}	NA
EVT-P-0016 960-T01-IM 6-100	P-0016 960-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr ee _pr ime	294478 38	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7453:EML4_c.3172+489:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017 592-T01-IM 5-101	P-0017 592-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr ee _pr ime	294475 48	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 1-5 fused with ALK exon 20-29) : c.494-5963:EML4_c.3172+779:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0029 400-T01-IM 6-102	P-0029 400-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294476 68	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 315+641:EML4_c.3172+65 9:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0004 218-T01-IM 5-103	P-0004 218-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 87	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 20 fused with ALK exon 20) : c.2068+156 :EML4_c.3173-693:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0017 978-T01-IM 6-104	P-0017 978-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7 306:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017 978-T02-IM 6-105	P-0017 978-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7 306:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0017 978-T03-IM 6-106	P-0017 978-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 09	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20-29): c.667+7 306:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018 242-T01-IM 6-107	P-0018 242-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294480 07	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused in-frame with ALK exons 20-29): c.668-7560:EML4_ c.3172+320:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018 280-T01-IM 6-108	P-0018 280-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294471 87	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exon 6 fused with ALK exon 20): c.494-6792: EML4_c.3173-793:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_j_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0003 913-T01-IM 3-109	P-0003 913-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294472 11	20	1058	True	retain ed	1.0	False	Protein kinase domai n			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0029 174-T01-IM 6-110	P-0029 174-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 56	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-19 fused with ALK exons 20-29): c.2 242+168:EML4_c.3173-26 2:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0018 969-T01-IM 6-112	P-0018 969-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294484 16	19	1028	False	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-760:EML4_c.3083:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019 491-T01-IM 6-113	P-0019 491-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 96	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 with ALK exons 20-29): c.1490- 486:EML4_c.3173-202:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019 513-T01-IM 6-114	P-0019 513-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294484 03	19	1032	False	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 19-29): c.1 489+2278:EML4_c.3096:A LKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0019 533-T01-IM 6-115	P-0019 533-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294464 94	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 with ALK exons 20-29): c.667+6837: EML4_c.3173-100:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0009 102-T01-IM 5-116	P-0009 102-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 13	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-12 fused in frame with ALK exons 20-29, which include the ALK kinase domain): c.149 0-244:EML4_c.3173-119:A LKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028 901-T01-IM 6-117	P-0028 901-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294466 07	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 316-844:EML4_c.3173-213 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0019 786-T01-IM 6-118	P-0019 786-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294475 75	19	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fused in-frame with ALK exons 20-29): c.2056+82:EML4_c.3172+752:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019 972-T01-IM 6-119	P-0019 972-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294481 01	19	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused to ALK exon 20-29) : c.493+1179:EML4_c.3172+226:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 124-T01-IM 6-120	P-0020 124-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294479 55	19	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused in-frame with ALK exons 20-29): c.1489+879:EML4_c.3172+372:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 138-T01-IM 6-121	P-0020 138-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294479 07	19	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-3016:EML4_c.3172+420:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003 383-T01-IM 5-122	P-0003 383-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294475 72	19	1058	True	retain ed	1.0	False	Protein kinase domai n			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0020 408-T01-IM 6-123	P-0020 408-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294467 79	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1489+1227:EML4_c.3173-385:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003 323-T01-IM 5-124	P-0003 323-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294475 30	19	1058	True	retain ed	1.0	False	Protein kinase domai n			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003 297-T01-IM 5-125	P-0003 297-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294482 59	19	1058	True	retain ed	1.0	False	Protein kinase domai n			NA Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0020 584-T01-IM 6-126	P-0020 584-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447048	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exon 20-29) : c.209-4846:EML4_c.3173-654:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 780-T01-IM 6-127	P-0020 780-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447084	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1 - 13 fused to ALK exons 20 - 29) : c.1489+1278:EML4_c.3173-690:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0020 825-T03-IM 6-128	P-0020 825-T03-IM6		ALK-EML4	EML4	in-frame	three_prime	29446767	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20-29):c.208+4890:EML4_c.3173-373:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0021 257-T01-IM 6-129	P-0021 257-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29447777	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-17 fused with ALK exons 20-29): c.1883-596:EML4_c.3172+550:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0003 206-T01-IM 5-130	P-0003 206-T01-IM5		ALK-EML4	EML4	unknown	three_prime	29448365	19	1045	False	retained	1.0	False	Protein kinase domain			NA Protein fusion: mid-exon (EML4-ALK)	NA
EVT-P-0021 495-T01-IM 6-131	P-0021 495-T01-IM6		ALK-EML4	EML4	in-frame	three_prime	29448313	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.493+2355:EML4_c.3172+14:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002 613-T01-IM 3-132	P-0002 613-T01-IM3		ALK-EML4	EML4	in-frame	three_prime	29447199	20	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0002 408-T02-IM 5-133	P-0002 408-T02-IM5		ALK-EML4	EML4	in-frame	three_prime	29448053	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 fuses with ALK exons 20-29) : c.2057-115:EML4_c.3172+274:ALKInv Protein fusion: in frame (EML4-ALK)	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ ro le	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mai ns	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0021 693-T01-IM 6-134	P-0021 693-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294470 02	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) Fusion (EML4 exons 1-14 fused with ALK exons 20-29): c.1 316-1509:EML4_c.3173-60 8:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 101-T01-IM 6-135	P-0022 101-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294477 59	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon18 fused with ALK exon 20): c.2057-510: EML4_c.2057-510:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 138-T01-IM 6-136	P-0022 138-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thre _pr ime	294465 84	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused with ALK exons 20-29): c.2 333:EML4_c.3173-190:AL Kinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0036 247-T01-IM 6-137	P-0036 247-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294478 57	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+107:EML4_c.3172+470: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036 106-T01-IM 6-138	P-0036 106-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294469 30	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) -ALK (NM_004304) fusion (EML4 exons 1-13 fused to ALK exons 20-29): c.1490- 2055:EML4_c.3173-536:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028 745-T01-IM 6-139	P-0028 745-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294475 09	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 523-2563:EML4_c.3172:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 398-T01-IM 6-140	P-0022 398-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294480 35	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.4 94-7853:EML4_c.3172+29 2:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0002 266-T01-IM 3-141	P-0002 266-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	five _pr ime	294472 88	20	1058	True	lost	0.0	False		Prot ein k inas e do mai n		ALK (NM_004304) and EML4 (NM_019063) reciprocal inversion (13026566bp): c.3173-889: ALK_c.208+1022:EML4inv Protein fusion: in frame (ALK-EML4)	NA
EVT-P-0002 266-T01-IM 3-142	P-0002 266-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294472 83	20	1058	True	retai ned	1.0	False	Protein kinase domai n			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028 541-T03-IM 6-143	P-0028 541-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294469 22	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.208+4631:EML4_c.3173- 528:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 926-T01-IM 6-144	P-0022 926-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294465 58	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29): c.4 93+983:EML4_c.3173-164: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0035 121-T01-IM 6-145	P-0035 121-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294473 83	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -4857:EML4_c.3172+944:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002 217-T02-IM 3-146	P-0002 217-T0 2-IM3		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294465 94	20	1058	True	retai ned	1.0	False	Protein kinase domai n			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0023 368-T01-IM 6-147	P-0023 368-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294467 13	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 20-29 including the kinase domain) : c.667+7004:EML 4_c.3173-319:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0002 024-T01-IM 3-148	P-0002 024-T0 1-IM3		ALK- EML 4	EML 4	in-fr ame	thre _pr ime	294471 33	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) inversion: c. 2242+31:EML4_c.3173-73 9:ALKinv Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0023 962-T01-IM 6-149	P-0023 962-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294488 20	19	1023	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused with ALK exons 19-29): c.667+4775:EML4_c.3068-389:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0024 136-T01-IM 6-150	P-0024 136-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294482 66	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-6 fused to ALK exons 20 -29): c.668-4498:EML4_c.3172+61:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0001 811-T01-IM 3-151	P-0001 811-T0 1-IM3		ALK- EML 4	EML 4	in-frame	three_prime	294482 05	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) Inversion: c.1315+2666:EML4_c.3172+122:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0028 541-T02-IM 6-152	P-0028 541-T0 2-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294469 22	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.208+4631:EML4_c.3173-528:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0024 907-T01-IM 6-153	P-0024 907-T0 1-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294468 77	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-2 fused to ALK exons 20 -29): c.209-990:EML4_c.3173-483:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034 569-T02-IM 6-154	P-0034 569-T0 2-IM6		ALK- EML 4	EML 4	in-frame	three_prime	294465 14	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243-217:EML4_c.3173-120:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034 450-T01-IM 6-155	P-0034 450-T0 1-IM6		ALK- EML 4	EML 4	unknown	three_prime	294463 54	20	1071	False	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon13 fused with ALK exons 20) : c.1641+555:EML4_c.3213:ALKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0001 532-T01-IM 3-156	P-0001 532-T0 1-IM3		ALK- EML 4	EML 4	in-frame	three_prime	294475 53	19	1058	True	retained	1.0	False	Protein kinase domain			NA Protein fusion: in frame (EML4-ALK)	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_ame	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0028 541-T01-IM 6-157	P-0028 541-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294469 22	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.2 08+4631:EML4_c.3173-52 8:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0026 407-T01-IM 6-158	P-0026 407-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294463 92	20	1059	False	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-18 fused with ALK exons 20-29): c.1 981-2450:EML4_c.3175:AL Kinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0026 650-T02-IM 6-159	P-0026 650-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	five _pr ime	294474 51	19	1058	True	lost	0.0	False		Prot ein k inas e do mai n		EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fuses with ALK exons 20-29) : c.6 67+7374:EML4_c.3172+87 6:ALKinv Protein Fusion: in frame {ALK:EML4}	NA
EVT-P-0026 657-T01-IM 6-160	P-0026 657-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294469 52	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fuses with ALK exons 20-29) : c.2 312:EML4_c.3173-558:AL Kinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0026 657-T03-IM 6-161	P-0026 657-T0 3-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294469 52	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-5 fused to ALK exons 20-29): c.2312: EML4_c.3173-558:ALKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0015 152-T02-IM 6-162	P-0015 152-T0 2-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294470 79	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused in-frame to ALK exons 20-29): c.2242+127:EML4_ c.3173-685:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0032 862-T01-IM 6-163	P-0032 862-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294474 88	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) Fusion(EML4 exon 11 fused to ALK exon20): c.1 489+671:EML4_c.3172+83 9:ALK Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0027 119-T03-IM 6-164	P-0027 119-T0 3-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294467 11	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 316-837:EML4_c.3173-317 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 683-T01-IM 6-165	P-0032 683-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294477 63	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2122:EML4_c.3172+564: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0028 339-T01-IM 6-166	P-0028 339-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294476 80	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-2 fused with ALK exons 20-29): c.2 08+1245:EML4_c.3172+64 7:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 518-T01-IM 6-167	P-0032 518-T0 1-IM6		ALK- EML 4	EML 4	unk now n	thr_ee _pr ime	294483 59	19	1047	False	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+2607:EML4_c.314 0:ALKinv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0027 853-T01-IM 6-168	P-0027 853-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294477 34	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20-29): c.2243- 157:EML4_c.3172+593:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 110-T01-IM 6-169	P-0015 110-T0 1-IM6		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 18	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exons 1-5 with ALK exons 20-29) : c.668-1726: EML4_c.3173-124:ALKinv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0003 446-T01-IM 5-170	P-0003 446-T0 1-IM5		ALK- EML 4	EML 4	in-fr ame	thr_ee _pr ime	294465 34	20	1058	True	retai ned	1.0	False	Protein kinase domai n			NA Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0004 516-T02-IM 5-171	P-0004 516-T0 2-IM5		ALK- GCC 2	GC C2	in-fr ame	thr_ee _pr ime	294465 26	20	1058	True	retai ned	1.0	False	Protein kinase domai n			GCC2 (NM_181453) - ALK (NM_004304) rearrangement : c.4448-12 30:GCC2_c.3173-132:ALKi nv Protein fusion: in frame (GCC2-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0035 712-T01-IM 6-172	P-0035 712-T0 1-IM6		ALK-H6PD	H6PD	unknown	five_prime	29462508	13	785	True	lost	0.0	False		Protein kinase domain		ALK (NM_004304) rearrangement: t(1;2)(p36.22;p23.2)(chr1:g.9314150::chr2:g.29462508) Antisense Fusion	NA
EVT-P-0018 541-T01-IM 6-174	P-0018 541-T0 1-IM6		ALK-LRP1B	LRP1B	in-frame	five_prime	29450289	17	972	True	lost	0.0	False		Protein kinase domain		ALK (NM_004304) - LRP1B (NM_018557) Rearrangement : c.2914+151:ALK_c.344-85984:LRP1Bdup Protein Fusion: in frame {ALK:LRP1B}	NA
EVT-P-0027 092-T01-IM 6-176	P-0027 092-T0 1-IM6		ALK-PLEKHH2	PLEKHH2	in-frame	three_prime	29446506	20	1058	True	retained	1.0	False	Protein kinase domain			PLEKHH2 (NM_172069) - ALK (NM_004304) rearrangement: c.502+545:PLEKHH2_c.3173-112:ALKinv Protein Fusion: in frame {PLEKHH2:ALK}	NA
EVT-P-0027 363-T02-IM 6-177	P-0027 363-T0 2-IM6		ALK-RANBP2	RANBP2	in-frame	three_prime	29447364	19	1058	True	retained	1.0	False	Protein kinase domain			RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1556:RANBP2_c.3172+963:ALKinv Protein Fusion: in frame {RANBP2:ALK}	NA
EVT-P-0068 865-T02-IM 7-178	P-0068 865-T0 2-IM7		ALK-RANBP2	RANBP2	in-frame	five_prime	29447373	19	1058	True	lost	0.0	False		Protein kinase domain		ALK (NM_004304) - RANBP2 (NM_006267) rearrangement: c.3172+954:ALK_c.2602+1532:RANBP2inv Protein Fusion: in frame {ALK:RANBP2}	NA
EVT-P-0008 843-T01-IM 5-179	P-0008 843-T0 1-IM5		ALK-RANBP2	RANBP2	in-frame	three_prime	29446488	20	1058	True	retained	1.0	False	Protein kinase domain			RANBP2 (NM_006267) - ALK (NM_004304) rearrangement: c.2603-1070:RANBP2_c.3173-94:ALKinv Protein fusion: in frame (RANBP2-ALK)	NA
EVT-P-0015 699-T01-IM 6-181	P-0015 699-T0 1-IM6		ALK-SPTBN1	SPTBN1	in-frame	three_prime	29447437	19	1058	True	retained	1.0	False	Protein kinase domain			SPTBN1 (NM_003128) - ALK (NM_004304) fusion (SPTBN1 exon 7 fused to ALK exon 20): c.764-1028: SPTBN1_c.3172+890:ALKinv Protein fusion: in frame (SPTBN1-ALK)	NA
EVT-P-0016 193-T05-IM 6-182	P-0016 193-T0 5-IM6		ALK-STRN	STRN	unknown	three_prime	29448326	19	1058	True	retained	1.0	False	Protein kinase domain			STRN (NM_003162) - ALK (NM_004304) fusion (STRN exons 1-3 fused with ALK exons 20-29): c.412+4919:STRN_c.3172+1:ALKdel Protein Fusion: mid-exon {STRN:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0016 193-T02-IM 6-183	P-0016 193-T0 2-IM6		ALK-STR N	STR N	unk now n	thr ee _pr ime	294483 26	19	1058	True	retai ned	1.0	False	Protein kinase domai n			ALK (NM_004304) - STRN (NM_003162) :c.3172+1:A LK_c.412+4919:STRNdel Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0016 193-T01-IM 6-184	P-0016 193-T0 1-IM6		ALK-STR N	STR N	unk now n	thr ee _pr ime	294483 26	19	1058	True	retai ned	1.0	False	Protein kinase domai n			ALK (NM_004304) - STRN (NM_003162) :c.3172+1:A LK_c.412+4919:STRNdel Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0041 438-T01-IM 6-185	P-0041 438-T0 1-IM6		ALK-TTC 27	TTC 27	out- of-fr ame	five _pr ime	294478 17	19	1058	True	lost	0.0	False		Prot ein k inas e do mai n		ALK (NM_004304) - TTC27 (NM_017735) rearrangement: c.3172+510:ALK_c.538-4165:TTC27inv Protein Fusion: out of frame {ALK:TTC27}	NA
EVT-P-0006 617-T01-IM 5-187	P-0006 617-T0 1-IM5		ASA P2-A LK	ASA P2	unk now n	five _pr ime	301427 53	1	223	True	lost	0.0	False		Prot ein k inas e do mai n		ASAP2 (NM_003887) - ALK (NM_004304) rearrangement event : c.127-20148:ASAP2_c.667+106:ALKdel Antisense fusion	NA
EVT-P-0062 994-T02-IM 7-188	P-0062 994-T0 2-IM7		CLT C-AL K	CLT C	in-fr ame	thr ee _pr ime	294496 24	18	1023	True	retai ned	1.0	False	Protein kinase domai n			CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29449624::chr17:g.57768350) Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0051 864-T01-IM 6-189	P-0051 864-T0 1-IM6		CLT C-AL K	CLT C	in-fr ame	thr ee _pr ime	294488 39	19	1023	True	retai ned	1.0	False	Protein kinase domai n			CLTC (NM_004859) - ALK (NM_004304) fusion: t(2;17)(p23.2;q23.1)(chr2:g.29448839::chr17:g.57770376) Protein Fusion: in frame {CLTC:ALK}	NA
EVT-P-0025 388-T01-IM 6-190	P-0025 388-T0 1-IM6		CTN NA1- ALK	CTN NA1	in-fr ame	thr ee _pr ime	294464 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n			CTNNA1 (NM_001903) - ALK (NM_004304) rearrangement: t(2;5)(p23.2;q31.2)(chr2:g.29446465::chr5:g.138259019) Protein Fusion: in frame {CTNNA1:ALK}	NA
EVT-P-0051 068-T01-IM 6-191	P-0051 068-T0 1-IM6		DIAP H2-A LK	DIA PH2	in-fr ame	thr ee _pr ime	294475 66	19	1058	True	retai ned	1.0	False	Protein kinase domai n			DIAPH2 (NM_006729) - ALK (NM_004304) fusion: t(2;X)(p23.2;q21.33)(chr2:g.29447566::chrX:g.96713620) Protein Fusion: in frame {DIAPH2:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0005 759-T02-IM 6-192	P-0005 759-T0 2-IM6		EMIL IN1- ALK	EMI LIN1	in-fr ame	thr ee _pr ime	294507 41	17	939	True	retai ned	1.0	False	Protein kinase domai n			EMILIN1 (NM_007046) - ALK (NM_004304) fusion: c.2575+44:EMILIN1_c.281 6-203:ALKinv Protein Fusion: in frame {EMILIN1:ALK}	NA
EVT-P-0046 083-T01-IM 6-193	P-0046 083-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294466 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+986:EML4_c.3173-271:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049 375-T01-IM 6-194	P-0049 375-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294467 32	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +1166:EML4_c.3173-338:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 646-T03-IM 6-195	P-0015 646-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294480 28	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.209 -3909:EML4_c.3172+299:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014 385-T03-IM 6-196	P-0014 385-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 76	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1890:EML4_c.3172+651 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0027 119-T04-IM 7-197	P-0027 119-T0 4-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294467 11	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-837:EML4_c.3173-317:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0057 391-T02-IM 6-198	P-0057 391-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 92	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -3018:EML4_c.3173-698:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0034 334-T01-IM 6-199	P-0034 334-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 70	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4909:EML4_c.3173-176:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0025 095-T01-IM 6-200	P-0025 095-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294473 21	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 fused to ALK exons 20 -29): c.2243- 278:EML4_c.3173-927:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gen_e	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0025 040-T01-IM 6-201	P-0025 040-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294471 52	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exon 12 fused to ALK exon 20) : c.1316-194 0:EML4_c.3173-758:ALKIn v Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0007 705-T02-IM 6-202	P-0007 705-T0 2-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294474 17	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.159 4:EML4_c.3172+910:ALKIn v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0023 375-T02-IM 7-203	P-0023 375-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294478 54	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2728:EML4_c.3172+473: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0035 085-T03-IM 6-204	P-0035 085-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 79	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +7361:EML4_c.3173-685:A LKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0001 766-T04-IM 6-205	P-0001 766-T0 4-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294487 67	19	1023	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +3962:EML4_c.3068-336:A LKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0068 467-T02-IM 7-206	P-0068 467-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294482 95	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1566:EML4_c.3172+32: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 184-T04-IM 6-207	P-0022 184-T0 4-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 35	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1386:EML4_c.3173-641: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0036 106-T03-IM 7-208	P-0036 106-T0 3-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294469 30	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2055:EML4_c.3173-536: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0022 184-T01-IM 6-209	P-0022 184-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 35	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion (EML4 exons 1-12 fused with ALK exons 20-29): c.1 315+1386:EML4_c.3173-6 41:ALKInv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ ro le	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0037 404-T01-IM 6-210	P-0037 404-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294478 72	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4963:EML4_c.3172+455: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 491-T01-IM 6-211	P-0037 491-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294467 86	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) Fusion (EML4 exon12 fused with ALK exon20): c.1490-1120: EML4_c.3173-392:ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 668-T01-IM 6-212	P-0037 668-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294476 31	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2742:EML4_c.3172+696: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 925-T01-IM 6-213	P-0037 925-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294465 22	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4634:EML4_c.3173-128:A LKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0037 982-T01-IM 6-214	P-0037 982-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294464 79	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-138:EML4_c.3173-85:AL KInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038 201-T03-IM 6-215	P-0038 201-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294477 48	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+989:EML4_c.3172+579: ALKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0038 772-T01-IM 6-216	P-0038 772-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294466 87	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.234 1+147:EML4_c.3173-293:A LKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 860-T02-IM 6-217	P-0043 860-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294469 31	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-980:EML4_c.3173-537:A LKInv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067 093-T02-IM 7-218	P-0067 093-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294473 03	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +1508:EML4_c.3173-909:A LKInv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ ro le	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mai ns	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0017 592-T04-IM 6-219	P-0017 592-T0 4-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294475 48	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -5963:EML4_c.3172+779:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 261-T01-IM 6-220	P-0040 261-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294471 03	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1340:EML4_c.3173-709: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0015 961-T07-IM 6-221	P-0015 961-T0 7-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294465 26	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -6666:EML4_c.3173-132:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 553-T01-IM 6-222	P-0040 553-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294470 55	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+30:EML4_c.3173-661:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 574-T01-IM 6-223	P-0040 574-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294477 38	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-1589: EML4_c.3172+589:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 576-T01-IM 6-224	P-0040 576-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294473 21	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exon5 fused with ALK exon20) : c.668-821:E ML4_c.3173-927:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 600-T01-IM 6-225	P-0040 600-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294476 88	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +1795:EML4_c.3172+639: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0032 683-T02-IM 7-226	P-0032 683-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294477 63	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2122:EML4_c.3172+564: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014 385-T06-IM 7-227	P-0014 385-T0 6-IM7		EML 4-AL K	EML 4	in-fr ame	thre e_ _pr ime	294476 76	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1890:EML4_c.3172+651 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0067 069-T01-IM 7-228	P-0067 069-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294481 69	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+2305:EML4_c.3172+158 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014 385-T02-IM 6-229	P-0014 385-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 76	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1890:EML4_c.3172+651 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041 967-T01-IM 6-230	P-0041 967-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294481 47	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+447:EML4_c.3172+180: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0041 967-T02-IM 7-231	P-0041 967-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294481 47	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+447:EML4_c.3172+180: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0067 048-T02-IM 7-232	P-0067 048-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294473 91	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +2216:EML4_c.3172+936: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0066 568-T02-IM 7-233	P-0066 568-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294480 72	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +3428:EML4_c.3172+255: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0014 310-T02-IM 6-234	P-0014 310-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294475 44	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +6128:EML4_c.3172+783: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 041-T01-IM 6-235	P-0043 041-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 97	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+1782:EML4_c.3173-203: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0013 299-T02-IM 6-236	P-0013 299-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 38	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) -ALK (NM_004304) Fusion(EML4 exon5 fused with ALK exon20) : c.667+ 1302:EML4_c.3173-644:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0012 913-T02-IM 6-237	P-0012 913-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294469 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.209 -4133:EML4_c.3173-571:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 350-T01-IM 6-238	P-0043 350-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294475 13	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -4017:EML4_c.3172+814:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0043 363-T01-IM 6-239	P-0043 363-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294481 01	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.225 4:EML4_c.3172+226:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0043 594-T01-IM 6-240	P-0043 594-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294483 81	19	1040	False	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.339 -2119:EML4_c.3118:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0069 396-T02-IM 7-241	P-0069 396-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294469 44	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -4308:EML4_c.3173-550:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069 535-T01-IM 7-242	P-0069 535-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294477 64	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +3829:EML4_c.3172+563: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0069 617-T01-IM 7-243	P-0069 617-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294480 17	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +7821:EML4_c.3172+310: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044 432-T01-IM 6-244	P-0044 432-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294483 44	19	1052	False	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.205 7-576:EML4_c.3155:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0005 743-T02-IM 5-245	P-0005 743-T0 2-IM5		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294476 69	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-20 with ALK exons 20-29) : c.2243- 181:EML4_c.3172+658:AL Kinv Protein fusion: in frame (EML4-ALK)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0045 222-T03-IM 7-246	P-0045 222-T0 3-IM7		EML 4-ALK K	EML 4	unknown	three_prime	294475 83	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2277:EML4_c.3172+744:ALKInv Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0045 243-T01-IM 6-247	P-0045 243-T0 1-IM6		EML 4-ALK K	EML 4	in-frame	three_prime	294482 98	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3064:EML4_c.3172+29:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0045 341-T01-IM 6-248	P-0045 341-T0 1-IM6		EML 4-ALK K	EML 4	in-frame	three_prime	294479 20	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-4434:EML4_c.3172+407:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065 469-T02-IM 7-249	P-0065 469-T0 2-IM7		EML 4-ALK K	EML 4	in-frame	three_prime	294465 65	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-1949:EML4_c.3173-171:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0065 210-T01-IM 7-250	P-0065 210-T0 1-IM7		EML 4-ALK K	EML 4	in-frame	three_prime	294480 76	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-2012:EML4_c.3172+251:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012 873-T03-IM 7-251	P-0012 873-T0 3-IM7		EML 4-ALK K	EML 4	in-frame	three_prime	294464 39	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1489+2231:EML4_c.3173-45:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064 716-T02-IM 7-252	P-0064 716-T0 2-IM7		EML 4-ALK K	EML 4	in-frame	three_prime	294477 39	19	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668-3669:EML4_c.3172+588:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 368-T04-IM 6-253	P-0031 368-T0 4-IM6		EML 4-ALK K	EML 4	in-frame	three_prime	294471 91	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.1490-1154:EML4_c.3173-797:ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0045 912-T02-IM 6-254	P-0045 912-T0 2-IM6		EML 4-ALK K	EML 4	out-of-frame	three_prime	294465 07	20	1058	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.2154+1011:EML4_c.3173-113:ALKinv Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0031 824-T01-IM 6-255	P-0031 824-T0 1-IM6		EML 4-ALK K	EML 4	in-frame	three_prime	294505 57	17	939	True	retained	1.0	False	Protein kinase domain			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208+2971:EML4_c.2816-19:ALKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0046 083-T02-IM 7-256	P-0046 083-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294466 65	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+986:EML4_c.3173-271:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0046 594-T01-IM 6-257	P-0046 594-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294464 45	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +2217:EML4_c.3173-51:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0040 732-T02-IM 6-258	P-0040 732-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294466 36	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1832:EML4_c.3173-242: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0047 355-T01-IM 6-259	P-0047 355-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294466 09	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +925:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0047 355-T02-IM 6-260	P-0047 355-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294466 09	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +925:EML4_c.3173-215:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0064 342-T01-IM 7-261	P-0064 342-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 34	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.208 +4042:EML4_c.3173-140:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0063 585-T01-IM 7-262	P-0063 585-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294468 84	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -2813:EML4_c.3173-490:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0012 873-T02-IM 6-263	P-0012 873-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294464 39	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+2231:EML4_c.3173-45:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0062 999-T02-IM 7-264	P-0062 999-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294470 91	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2055:EML4_c.3173-697: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049 318-T01-IM 6-265	P-0049 318-T0 1-IM6		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294463 59	20	1070	False	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.215 4+2335:EML4_c.3208:ALKi nv Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0046 720-T01-IM 6-266	P-0046 720-T0 1-IM6		EML 4-AL K	EML 4	out-of-fr ame	thr_ee _pr ime	294466 69	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.164 1+1:EML4_c.3173-275:AL Kinv Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0049 375-T02-IM 6-267	P-0049 375-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294467 32	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +1166:EML4_c.3173-338:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0049 511-T01-IM 6-268	P-0049 511-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294469 91	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.224 2+9:EML4_c.3173-597:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0010 177-T02-IM 7-269	P-0010 177-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294473 58	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-2649:EML4_c.3173-964: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 827-T03-IM 6-270	P-0031 827-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294466 46	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-639:EML4_c.3173-252:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051 231-T01-IM 6-271	P-0051 231-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294467 74	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4130:EML4_c.3173-380:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0051 231-T03-IM 6-272	P-0051 231-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294467 74	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4130:EML4_c.3173-380:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0062 985-T02-IM 7-273	P-0062 985-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294471 99	20	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.667 +4561:EML4_c.3173-805:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0052 035-T03-IM 6-274	P-0052 035-T0 3-IM6		EML 4-AL K	EML 4	in-fr ame	thr_ee _pr ime	294476 56	19	1058	True	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.224 2+79:EML4_c.3172+671:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0061 297-T03-IM 7-275	P-0061 297-T0 3-IM7		EML 4-AL K	EML 4	unk now n	thr_ee _pr ime	294483 75	19	1042	False	retain ed	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+931:EML4_c.3124:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_j_d	fusio_n_name	part_ner_gen_e	fra_me_sta_tus	tar_get_ro_le	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_sta_tus	domain_r_etained_fr_action	domain_is_tru_nicated	retain_ed_do_mains	lost_do_mains	disrupt_ed_do_mains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0052 297-T02-IM 6-276	P-0052 297-T0 2-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294482 55	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-161:EML4_c.3172+72:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0061 297-T02-IM 7-277	P-0061 297-T0 2-IM7		EML 4-AL K	EML 4	unk now n	thr ee _pr ime	294483 75	19	1042	False	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+931:EML4_c.3124:ALKin v Protein Fusion: mid-exon {EML4:ALK}	NA
EVT-P-0005 786-T01-IM 5-278	P-0005 786-T0 1-IM5		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294482 39	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-18 with ALK exons 20-29) : c.2057- 186:EML4_c.3172+88:ALKi nv Protein fusion: in frame (EML4-ALK)	NA
EVT-P-0060 148-T01-IM 7-279	P-0060 148-T0 1-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294477 57	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.148 9+2600:EML4_c.3172+570 :ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0059 699-T03-IM 7-280	P-0059 699-T0 3-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294474 67	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1990:EML4_c.3172+860: ALKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0031 827-T06-IM 7-281	P-0031 827-T0 6-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294466 46	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-639:EML4_c.3173-252:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054 263-T01-IM 6-282	P-0054 263-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294465 57	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -210:EML4_c.3173-163:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0059 257-T02-IM 7-283	P-0059 257-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294482 10	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -982:EML4_c.3172+117:AL Kinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0054 408-T01-IM 6-284	P-0054 408-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee _pr ime	294469 02	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.224 3-262:EML4_c.3173-508:A LKinv Protein Fusion: in frame {EML4:ALK}	NA

event_id	sampl e_id	pa tie nt _i d	fusio n_n ame	part ner_ gene	fra me_ sta tus	tar get_ role	breakp oint_ge nomic	break point_ exon	breakpoint _protein_p osition	is_intron ic_break point	dom ain_ statu s	domain_r etained_fr action	domain _is_trun cated	retain ed_do mains	lost_ do mains	disrupt ed_do mains	source_annotation_text	source_site2 _effect_on_f rame
EVT-P-0008 332-T02-IM 6-285	P-0008 332-T0 2-IM6		EML 4-AL K	EML 4	unk now n	thr ee_ _pr ime	294469 13	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_001145076) - ALK (NM_004304) fusion: c.1315+828:EML4_c.3173- 519:ALKdup Antisense Fusion	NA
EVT-P-0058 599-T02-IM 7-286	P-0058 599-T0 2-IM7		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294464 15	20	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.149 0-1345:EML4_c.3173-21:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0044 042-T01-IM 6-287	P-0044 042-T0 1-IM6		EML 4-AL K	EML 4	in-fr ame	thr ee_ _pr ime	294477 27	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion: c.668 -1681:EML4_c.3172+600:A LKinv Protein Fusion: in frame {EML4:ALK}	NA
EVT-P-0019 560-T01-IM 6-288	P-0019 560-T0 1-IM6		EML 4-AL K	EML 4	out- of-fr ame	thr ee_ _pr ime	294479 52	19	1058	True	retai ned	1.0	False	Protein kinase domai n			EML4 (NM_019063) - ALK (NM_004304) fusion (EML4 exons 1-16 with ALK exons 20-29): c.1968- 2879:EML4_c.3172+375:A LKinv Protein Fusion: out of frame {EML4:ALK}	NA
EVT-P-0044 526-T02-IM 6-289	P-0044 526-T0 2-IM6		FN1- ALK	FN1	in-fr ame	thr ee_ _pr ime	294485 33	19	1023	True	retai ned	1.0	False	Protein kinase domai n			FN1 (NM_212482) - ALK (NM_004304) fusion: c.181 9+332:FN1_c.3068-102:AL Kdel Protein Fusion: in frame {FN1:ALK}	NA
EVT-P-0030 808-T01-IM 6-290	P-0030 808-T0 1-IM6		HMB OX1- ALK	HM BOX 1	unk now n	thr ee_ _pr ime	294477 13	19	1058	True	retai ned	1.0	False	Protein kinase domai n			HMBBOX1 (NM_001135726) ALK (NM_004304) Rearrangement : t(2;8)(p23 .3,p21.1)(chr2:g.29447713: :chr8:g.28904937) Protein Fusion: mid-exon {HMBBOX1:ALK}	NA
EVT-P-0062 804-T03-IM 7-291	P-0062 804-T0 3-IM7		KIF5 B-AL K	KIF5 B	in-fr ame	thr ee_ _pr ime	294477 91	19	1058	True	retai ned	1.0	False	Protein kinase domai n			KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)p23.2;p11.22)(chr2:g.294 47791::chr10:g.32304833) Protein Fusion: in frame {KIF5B:ALK}	NA
EVT-P-0005 711-T02-IM 6-292	P-0005 711-T0 2-IM6		KIF5 B-AL K	KIF5 B	in-fr ame	thr ee_ _pr ime	294495 48	18	1023	True	retai ned	1.0	False	Protein kinase domai n			KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2 ;10)(p23.2;p11.22)(chr2:g.2 9449548::chr10:g.3230476 5) Protein fusion: in frame (KIF5B-ALK)	NA

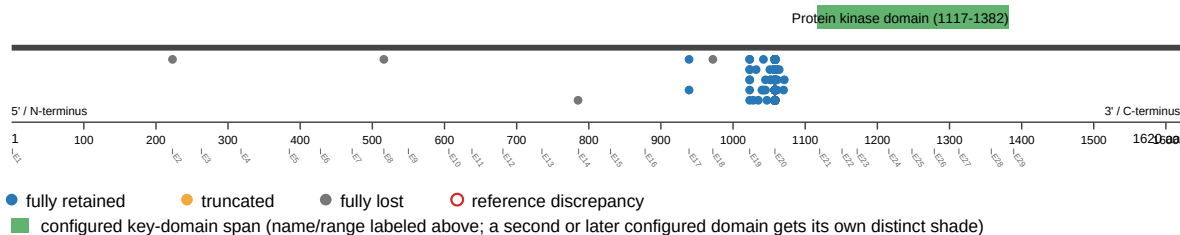
event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0008 764-T01-IM 5-293	P-0008 764-T0 1-IM5		KIF5 B-AL K	KIF5 B	unk now n	thr ee _pr ime	294483 27	19	1058	False	retai ned	1.0	False	Protein kinase domai n			KIF5B (NM_004521) - ALK (NM_004304) fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750) Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0017 924-T01-IM 6-294	P-0017 924-T0 1-IM6		KIF5 B-AL K	KIF5 B	in-fr ame	thr ee _pr ime	294479 45	19	1058	True	retai ned	1.0	False	Protein kinase domai n			KIF5B (NM_004521) - ALK (NM_004304) fusion: t(2;10)(p23.2;p11.22)(chr2:g.29447945::chr10:g.32305310) Protein Fusion: in frame {KIF5B:ALK}	NA
EVT-P-0008 764-T02-IM 6-295	P-0008 764-T0 2-IM6		KIF5 B-AL K	KIF5 B	unk now n	thr ee _pr ime	294483 27	19	1058	False	retai ned	1.0	False	Protein kinase domai n			KIF5B (NM_004521) - ALK (NM_004304) Fusion (KIF5B exons 1-24 fused with ALK exons 19-29) : t(2;10)(p23.2;p11.22)(chr2:g.29448327::chr10:g.32305750) Protein fusion: mid-exon (KIF5B-ALK)	NA
EVT-P-0067 512-T01-IM 7-297	P-0067 512-T0 1-IM7		NOT CH1- ALK	NOT CH1	in-fr ame	thr ee _pr ime	294491 52	18	1023	True	retai ned	1.0	False	Protein kinase domai n			NOTCH1 (NM_017617) - ALK (NM_004304) fusion: t(2;9)(p23.2;q34.3)(chr2:g.29449152::chr9:g.139416748) Protein Fusion: in frame {NOTCH1:ALK}	NA
EVT-P-0034 044-T01-IM 6-298	P-0034 044-T0 1-IM6		NRP 2-AL K	NRP 2	in-fr ame	thr ee _pr ime	294485 50	19	1023	True	retai ned	1.0	False	Protein kinase domai n			NRP2 (NM_201266) - ALK (NM_004304) rearrangement: c.1291+575_c.3068-119inv Protein Fusion: in frame {NRP2:ALK}	NA
EVT-P-0055 667-T06-IM 7-299	P-0055 667-T0 6-IM7		PICA LM- ALK	PIC ALM	unk now n	thr ee _pr ime	294463 86	20	1061	False	retai ned	1.0	False	Protein kinase domai n			PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0055 667-T02-IM 6-300	P-0055 667-T0 2-IM6		PICA LM- ALK	PIC ALM	unk now n	thr ee _pr ime	294463 86	20	1061	False	retai ned	1.0	False	Protein kinase domai n			PICALM (NM_007166) - ALK (NM_004304) rearrangement: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) Protein Fusion: mid-exon {PICALM:ALK}	NA
EVT-P-0055 667-T04-IM 6-301	P-0055 667-T0 4-IM6		PICA LM- ALK	PIC ALM	unk now n	thr ee _pr ime	294463 86	20	1061	False	retai ned	1.0	False	Protein kinase domai n			PICALM (NM_007166) - ALK (NM_004304) fusion: t(2;11)(p23.2;q14.2)(chr2:g.29446386::chr11:g.85681391) Protein Fusion: mid-exon {PICALM:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_truncated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0054 253-T01-IM 6-302	P-0054 253-T0 1-IM6		PLE KHA 7-AL K	PLE KHA 7	unk now n	thr ee _pr ime	294483 96	19	1035	False	retai ned	1.0	False	Protein kinase domai n			PLEKHA7 (NM_175058) - ALK (NM_004304) fusion: t (2;11)(p23.2;p15.1)(chr2:g. 29448396::chr11:g.168138 09) Protein Fusion: mid-exon {PLEKHA7:ALK}	NA
EVT-P-0059 884-T01-IM 7-303	P-0059 884-T0 1-IM7		PPP 1CB- ALK	PPP 1CB	in-fr ame	thr ee _pr ime	294474 41	19	1058	True	retai ned	1.0	False	Protein kinase domai n			PPP1CB (NM_206876) - ALK (NM_004304) fusion: c.52+5708:PPP1CB_c.317 2+886:ALKinv Protein Fusion: in frame {PPP1CB:ALK}	NA
EVT-P-0048 893-T01-IM 6-307	P-0048 893-T0 1-IM6		SPT BN1- ALK	SPT BN1	unk now n	thr ee _pr ime	294483 46	19	1051	False	retai ned	1.0	False	Protein kinase domai n			SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.763+1517:SPTBN1_c.31 53:ALKinv Protein Fusion: mid-exon {SPTBN1:ALK}	NA
EVT-P-0053 963-T02-IM 6-308	P-0053 963-T0 2-IM6		SPT BN1- ALK	SPT BN1	unk now n	thr ee _pr ime	294483 45	19	1052	False	retai ned	1.0	False	Protein kinase domai n			SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-1123:SPTBN1_c.315 4:ALKinv Protein Fusion: mid-exon {SPTBN1:ALK}	NA
EVT-P-0045 724-T01-IM 6-309	P-0045 724-T0 1-IM6		SPT BN1- ALK	SPT BN1	in-fr ame	thr ee _pr ime	294467 96	20	1058	True	retai ned	1.0	False	Protein kinase domai n			SPTBN1 (NM_003128) - ALK (NM_004304) fusion: c.764-456:SPTBN1_c.3173 -402:ALKinv Protein Fusion: in frame {SPTBN1:ALK}	NA
EVT-P-0042 596-T02-IM 6-310	P-0042 596-T0 2-IM6		SQS TM1- ALK	SQS TM1	in-fr ame	thr ee _pr ime	294465 11	20	1058	True	retai ned	1.0	False	Protein kinase domai n			SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t (2;5)(p23.2;q35.3)(chr2:g.2 9446511::chr5:g.17925460 3) Protein Fusion: in frame {SQSTM1:ALK}	NA
EVT-P-0042 596-T01-IM 6-311	P-0042 596-T0 1-IM6		SQS TM1- ALK	SQS TM1	in-fr ame	thr ee _pr ime	294465 11	20	1058	True	retai ned	1.0	False	Protein kinase domai n			SQSTM1 (NM_003900) - ALK (NM_004304) fusion: t (2;5)(p23.2;q35.3)(chr2:g.2 9446511::chr5:g.17925460 3) Protein Fusion: in frame {SQSTM1:ALK}	NA
EVT-P-0053 901-T01-IM 6-312	P-0053 901-T0 1-IM6		STR N-AL K	STR N	unk now n	thr ee _pr ime	294463 75	20	1064	False	retai ned	1.0	False	Protein kinase domai n			STRN (NM_003162) - ALK (NM_004304) fusion: c.339 -2565:STRN_c.3192:ALKd el Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0048 102-T03-IM 6-313	P-0048 102-T0 3-IM6		STR N-AL K	STR N	in-fr ame	thr ee _pr ime	294470 56	20	1058	True	retai ned	1.0	False	Protein kinase domai n			STRN (NM_003162) - ALK (NM_004304) fusion: c.412 +2761:STRN_c.3173-662: ALKdel Protein Fusion: in frame {STRN:ALK}	NA

event_id	sampl_e_id	pa_tie_nt_id	fusio_n_name	part_ner_gene	fra_me_status	tar_get_role	breakp_o_int_ge_nomic	break_point_exon	breakpoint_protein_p_osition	is_intron_ic_break_point	dom_ain_status	domain_r_etained_fr_action	domain_is_trun_cated	retain_ed_domains	lost_do_mains	disrupt_ed_domains	source_annotation_text	source_site2_effect_on_f_rame
EVT-P-0033 066-T01-IM 6-314	P-0033 066-T0 1-IM6		STR N-AL K	STR N	unk now n	thr ee _pr ime	294483 65	19	1045	False	retai ned	1.0	False	Protein kinase domai n			STRN (NM_003162) - ALK (NM_004304) fusion(STRN exons 1-3 fused to ALK exons 19-29): c.413-1084:STRN_c.3134:ALKdel Protein Fusion: mid-exon {STRN:ALK}	NA
EVT-P-0048 102-T02-IM 6-315	P-0048 102-T0 2-IM6		STR N-AL K	STR N	in-fr ame	thr ee _pr ime	294470 56	20	1058	True	retai ned	1.0	False	Protein kinase domai n			STRN (NM_003162) - ALK (NM_004304) fusion: c.412+2761:STRN_c.3173-662: ALKdel Protein Fusion: in frame {STRN:ALK}	NA
EVT-P-0056 097-T01-IM 6-316	P-0056 097-T0 1-IM6		TFG- ALK	TFG	in-fr ame	thr ee _pr ime	294478 96	19	1058	True	retai ned	1.0	False	Protein kinase domai n			TFG (NM_001195478) - ALK (NM_004304) rearrangement: t(2;3)(p23.2;q12.2)(chr2:g.29447896::chr3:g.100456163) Protein Fusion: in frame {TFG:ALK}	NA
EVT-P-0045 363-T02-IM 6-318	P-0045 363-T0 2-IM6		TPM 1-AL K	TPM 1	in-fr ame	thr ee _pr ime	294465 14	20	1058	True	retai ned	1.0	False	Protein kinase domai n			TPM1 (NM_001018020) - ALK (NM_004304) fusion: t (2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833) Protein Fusion: in frame {TPM1:ALK}	NA
EVT-P-0045 363-T01-IM 6-319	P-0045 363-T0 1-IM6		TPM 1-AL K	TPM 1	in-fr ame	thr ee _pr ime	294465 14	20	1058	True	retai ned	1.0	False	Protein kinase domai n			TPM1 (NM_001018020) - ALK (NM_004304) fusion: t (2;15)(p23.2;q22.2)(chr2:g.29446514::chr15:g.63358833) Protein Fusion: in frame {TPM1:ALK}	NA
EVT-P-0053 383-T01-IM 6-320	P-0053 383-T0 1-IM6		TRA F3-A LK	TRA F3	in-fr ame	thr ee _pr ime	294476 88	19	1058	True	retai ned	1.0	False	Protein kinase domai n			TRAF3 (NM_003300) - ALK (NM_004304) fusion: t (2;14)(p23.2;q32.32)(chr2:g.29447688::chr14:g.103370282) Protein Fusion: in frame {TRAF3:ALK}	NA
EVT-P-0001 737-T01-IM 3-321	P-0001 737-T0 1-IM3		VCL- ALK	VCL	in-fr ame	thr ee _pr ime	294473 90	19	1058	True	retai ned	1.0	False	Protein kinase domai n			VCL (NM_014000) - ALK (NM_004304) Translocation : t(10;2)(q22.2;p23.2)(chr10:c.2435-486::chr2:c.3172+937) Protein fusion: in frame (VCL-ALK)	NA
EVT-P-0045 566-T01-IM 6-322	P-0045 566-T0 1-IM6		ZFP M2- ALK	ZFP M2	in-fr ame	thr ee _pr ime	294479 02	19	1058	True	retai ned	1.0	False	Protein kinase domai n			ZFPM2 (NM_012082) - ALK (NM_004304) fusion: t (2;8)(p23.2;q23.1)(chr2:g.29447902::chr8:g.106384092) Protein Fusion: in frame {ZFPM2:ALK}	NA

domain retention outliers

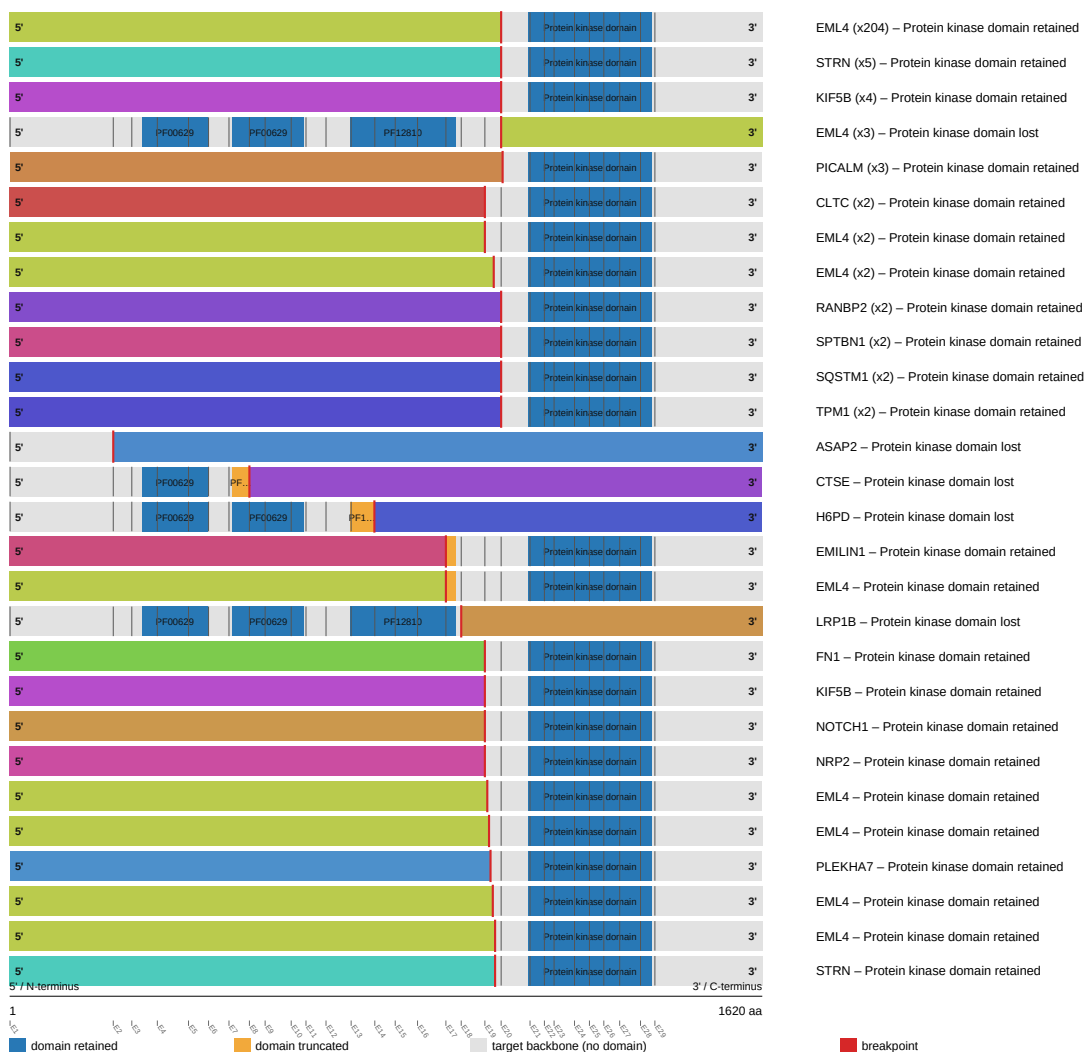
ALK domain-retention track



fusion schematic

ALK fusion-transcript schematic

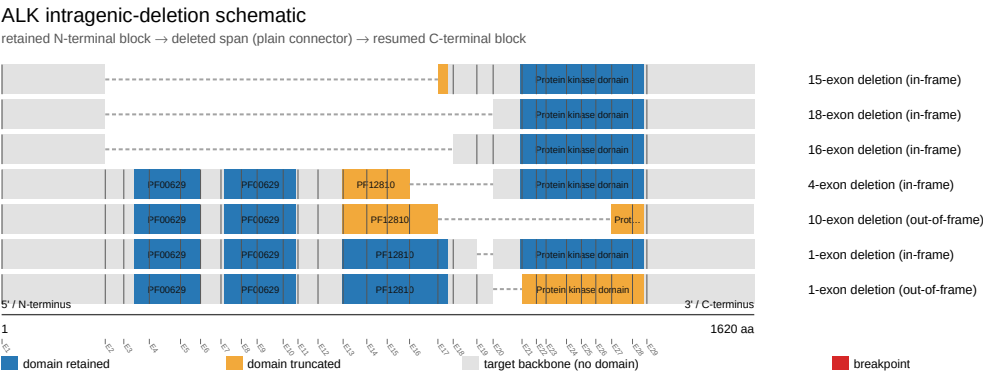
Fusion order varies by row; follow each row's 5' and 3' labels (ALK portion is domain-colored, exon ticks)



Partner-block colors are decorative; row labels identify partners.

Showing the top 28 of 50 partner/breakpoint groups by recurrence; see results.tsv for the complete list.

intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

